



DEALER BEST PRACTICE SERIES

Wheel Station Upender Equipment

Application Maintenance Site Management **Component Rebuild** Component Life Management Safety MARC Management

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1.0 Introduction

Handling large wheel groups during rebuild can be a difficult task. A wheel group typically arrives to the CRC mounted on a shipping stand. It must then be rotated from the horizontal position to a vertical position before disassembly. After rebuild, it must be again rotated from the vertical position back to a horizontal position for mounting on a shipping stand.

The process normally involves the use of two overhead cranes and multiple chains to perform the necessary rotations. It also requires several technicians to perform the operation.

Industrial equipment commonly called “upenders” are manufactured by several suppliers and are well-suited to the task of efficiently and securely rotating heavy components from horizontal to vertical positions and back.



Horizontal shipping position



Disassemble / assemble position

2.0 Best Practice Description

Industrial “upenders” are manufactured by a number of suppliers, such as Bushman and Titan. They are available in sizes ranging up to 30+ ton capacity. They typically can move a wheel station from horizontal to vertical position in less than 30 seconds. Some dealers have fabricated their own wheel group upender tooling.

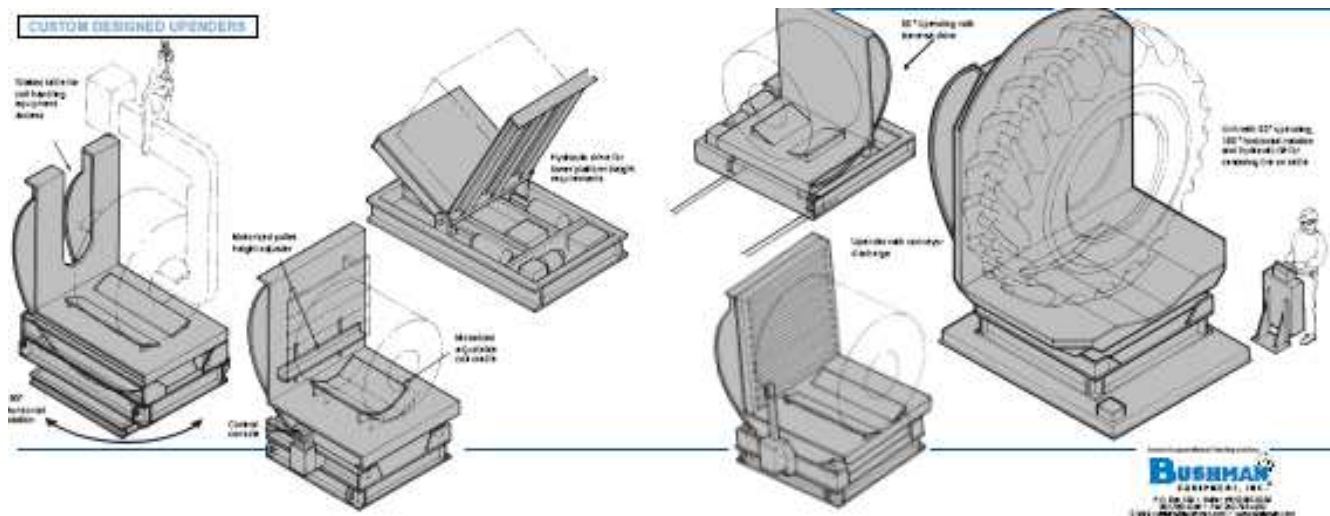
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Wheel Station Upender Equipment

DATE
4/11/2017

CHG
NO
02

NUMBER
0707-4.4-1090



TITAN UPENDERS

Autolap Titan Upenders reposition or turn upright a wide variety of heavy, bulky objects with minimal floor space requirements. They may be used in conjunction with automated coil cart systems or alone.

18 Models,
Maximum
Capacity 15,000
lbs. Can Be
Specified With
Lifts, Customizing
For Specific
Applications.

Lubricated
for life
composite
bearings

Long life
fabrication

TYPICAL MARKETS

Metal Working
Paper Mills
Foundries
Automotive Plants
Machine Shops

GENERAL APPLICATIONS

Coil Handling/Reorientation
Die Handling
Engine Block Manufacturing/Repair
Coil Palletizing
Roll Handling

90-degree tilters, flip-
top tilters and special
high-hinge tilters are
also available.



Examples of commercially available upenders

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Upenders installed in Cat dealer CRC's

3.0 Implementation Steps

It is best to work with a reputable supplier to identify proper requirements before installing an upender. You will need to know the maximum forecast load weight and floor space available in the wheel group rebuild area.

Some upenders have power options, such as electric or hydraulic. It is important that an upender used for wheel group handling has both power up and power down.

4.0 Benefits

- Raising and lowering wheel groups takes 50%+ less time.
- Much reduced risk of damaging the wheel group during handling.
- Requires only one technician to operate.
- Frees up overhead cranes for other tasks.
- Lesser risk of accidents.

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5.0 Resources Required

Costs vary by options and capacity. Contact a supplier for a quote. An investment analysis should be preformed to measure cost versus benefit.

Suppliers generally work through regional representatives and distributors. Contact information is available on their websites. See also:

Titan Worldwide Inc.
<http://www.titanworldwide.com/>

Bushman Equipment Inc.
<http://www.bushman.com/>

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

8.0 Acknowledgements

This Best Practice was written by:

Dale Brehm
Caterpillar Global Mining
6 Sigma Black Belt

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